

Detecting North Atlantic Right Whale Vocalizations in the Southeast US Calving Grounds During the Winter Calving Season Using Bottom-Mounted Hydrophones and a Glider

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The critically endangered North Atlantic right whale (NARW) migrates seasonally from its northern foraging grounds along the east coast of North America to its southeast US calving grounds. Right whale mothers typically give birth to their calves off the coast of South Carolina, Georgia, or Florida during the winter period, between December and March. Passive acoustic monitoring (PAM) is a common method for detecting the presence of baleen whales, such as the NARW, using hydrophones to detect low-frequency calls. NARWs migrate in shallow coastal waters and tend to reduce vocalizations when a newborn calf is present, and both of these factors reduce their acoustic detectability during this migration. To inform the design of effective PAM deployment configurations in this region, three bottom-mounted, archival hydrophones were deployed at increasing depths (15.9m, 19.5m, 22m) and increasing distances off the coast of Savannah, Georgia, during the NARW calving season. A Slocum Glider equipped with a digital acoustic monitoring instrument (DMON2) to enable near real-time capability to transmit acoustic data every 4 hours via Iridium satellite communication was deployed concurrently. The glider was constrained to operate in depths greater than 10 m of water, just at the edge of the designated right whale critical habitat. A pilot glider-based PAM program was started in 2023. So far, this program has consisted of a 2-week mission in January 2023, two 4-week missions in January through March 2024, and two missions in 2025, one in January and a shorter one in February. During these missions, there were 16 possible and 3 definite right whale acoustic detections. A detection on January 20, 2024 was the first definite glider-based acoustic detection in the Southeast US, eliciting near real-time detection communication via WhaleMap, Whale Alert and Robots4Whales, as well as the Early Warning System notifications program unique to the Southeast US. These pilot missions indicate that glider-based PAM may be a useful tool in supplementing aerial-based detections of right whales in the calving ground, especially providing coverage when aerial surveillance is not feasible due to weather or other logistical constraints and filling gaps in space and time.